

GEOMETRY & SPATIAL ANALYTICS

Honors Syllabus & Curricular Roadmap — Academic Year 2026–2027

This course approaches Euclidean geometry as a structural system of logical deduction, transformation mechanics, and coordinate analytics. Students will move systematically from spatial patterns to rigorous axiomatic proofs and classical straightedge-and-compass architectural constructions. The curriculum bridges pure mathematics with hands-on geometric modeling.

COURSE PARAMETERS & FRAMEWORKS

ONE-WEEK ROLLING DEADLINE FRAMEWORK

To promote conceptual mastery and structured pacing, all course work follows a strict **one-week rolling deadline**. Complete solutions to textbook sections, coordinate matrices, and proof packages are due exactly seven calendar days following their formal assignment in class.

DRAFTING & PRECISION STANDARDS

All graphical coordinate systems and multi-step proofs must be documented within a dedicated engineering grid journal. Classical construction sequences require verified drawing tools, including a mechanical locking compass, a clear straightedge, and a standard protractor.

Master Curricular Roadmap — Semester 1

TIMELINE	UNIT	CORE INSTRUCTIONAL FOCUS	ASSIGNED PROBLEM SETS & TASKS
Aug 31	Intro	Course Overview & Axiomatic Systems	Review Intro Slides & Geometry Foundations
Sep 1 – Sep 2	1.A	Points, Lines, Planes, and Segment Measurement	Textbook pg. 9: #7-8, 13, 16, 17, 25, 31-33; pg. 17: #1-4, 6-10, 14-15, 21-23
Sep 3 – Sep 4	1.B	Angular Measurement Matrices & Postulates	Textbook pg. 24: #2, 4-8, 15-17, 27, 29-30
Sep 8 – Sep 9	1.C	Geometric Pairs of Angles & Properties	Textbook pg. 31: #2, 7-8, 11, 13-17, 25-27, 29-30, 34-37
Sep 10 – Sep 11	1.D	Coordinate Plane Midpoint & Distance Formulas	Textbook pg. 47: #1, 7-9, 12-16, 19-20, 26-27
Sep 14 – Sep 15	Rev	Module 1 Unified Concept Review	Textbook Review pg. 60: #4-14, 16-22, 32-33, 35-36

Sep 16	Test	MODULE 1 SUMMATIVE EXAMINATION	In-Class Evaluation: Foundations & Coordinate Spaces
Sep 17 – Sep 21	2.A	Parallel & Perpendicular Lines: Algebraic Proofs	Complete Algebraic Proof Worksheets packet #1
Sep 22 – Sep 23	2.B1	Geometric Proof Structures with Lines	Complete Geometric Proof Worksheet #2
Sep 24 – Sep 25	2.B2	Geometric Proof Structures with Angles	Complete Angular Proof Worksheet #3
Sep 28	PIR	<i>No Class — Staff Professional Development Day</i>	—
Sep 29 – Sep 30	2.C	Lines, Angles, & Transversals Frameworks	Performance Quiz 1 & Proof Worksheet #4
Oct 1 – Oct 2	2.D	Proving Systems of Parallel Lines	Complete Proof Worksheet #5
Oct 5 – Oct 6	2.E	Slope & Linear Equations in Coordinate Planes	Complete Coordinate Proof Worksheet #6
Oct 7 – Oct 8	Rev	Module 2 Synthesis & Target Review	Complete Module 2 Review Packet Components
Oct 9	Test	MODULE 2 SUMMATIVE EXAMINATION	In-Class Evaluation: Line Proofs & Transversal Systems

TIMELINE	UNIT	CORE INSTRUCTIONAL FOCUS	ASSIGNED PROBLEM SETS & TASKS
Oct 13 – Oct 14	3.A	Triangle Congruence: Angle & Structural Classifications	Textbook pg. 219: #3-10; pg. 227: #5-6, 9-10, 12-13, 19, 22
Oct 15 – Oct 16	PIR	<i>No Class — State Professional Development Break</i>	—
Oct 19 – Oct 20	3.B	Congruent Triangles & Geometric Correspondence Mapping	Complete 3.B Curricular Worksheet Packet
Oct 21 – Oct 22	3.C	Congruence Theorems (SSS, SAS, ASA, AAS, HL Proofs)	Textbook pg. 246: #8, 11-12, 14-18; pg. 256: #4-5, 16-17, 22, 27-28
Oct 23 – Oct 26	3.D	Analytical Application of CPCTC within Proof Networks	Textbook pg. 262: #7-9, 17-19, 24, 28
Oct 27 – Oct 28	3.E	Isosceles and Equilateral Triangle Postulates	Textbook pg. 276: #13-15, 17-18, 22-25, 33-34
Oct 29 – Oct 30	Rev	Module 3 Part I Conceptual Synthesis	Complete Module 3 Comprehensive Review 1 Guide
Nov 2 – Nov 3	Test	MODULE 3 PART I TESTING MATRIX	In-Class Evaluation: Triangle Congruence Frameworks
Nov 4 – Nov 5	3.F	Midsegment Theorems & Coordinate Analytics	Textbook pg. 324: #11-16, 19-25 odd, 30-35
Nov 6 – Nov 9	3.G	Triangle Inequalities & Boundary Constraints	Textbook pg. 336: #18-19, 20-25, 34-35, 42-47
Nov 10 – Nov 11	3.H	Pythagorean Theorem, Radical Simplification, & Inequalities	Textbook pg. 352: #19-27, 30-32
Nov 12 – Nov 13	Rev	Module 3 Part II Conceptual Synthesis	Complete Module 3 Review Guide 2 Packets
Nov 16 – Nov 17	Test	MODULE 3 PART II TESTING MATRIX	In-Class Evaluation: Advanced Triangles & Inequalities
Nov 18 – Nov 20	4.A	Quadrilaterals: Interior & Exterior Polygon Properties	Textbook pg. 386: #16-25, 29-36, 51 & 52 (PIR Days overlap)
Nov 23 – Nov 24	4.B	Parallelograms, Rectangles, Rhombuses, and Squares	Textbook pg. 395: #16-24 even, 27-30, 41, 42, 46, 47; pg. 412: #10-13, 18-23 odd, 24-28
Nov 25 – Nov 27	Break	<i>No Class — Thanksgiving Recess</i>	—

Nov 30 – Dec 1	4.C	Analytical Geometric Properties of Kites and Trapezoids	Textbook pg. 432: #14-22, 28-32 even, 34-36
Dec 2 – Dec 3	4.D	Sufficient Conditions for Parallelograms & Review	Textbook pg. 438: #8-14 even, 15-25 odd, 35-47 odd, 50-54 even
Dec 4	Test	MODULE 4 SUMMATIVE EXAMINATION	In-Class Evaluation: Quadrilaterals & Polygon Systems
Dec 8 – Dec 9	5.A	Transformations: Grid Reflections & Translations	Textbook pg. 827: #13-18, 20-23; pg. 834: #11-14, 17-19, 29-34
Dec 10 – Dec 11	5.B	Symmetry & Rotational Mapping Rules	Textbook pg. 842: #11-15, 18-21, 27-30; pg. 859: #13-18, 38-43
Dec 14 – Dec 15	5.C	Composite Coordinate Grid Transformations & Synthesis	Launch Spatial Design Transformation Project Module
Dec 16 – Dec 18	Proj	TRANSFORMATION PROJECT COMPONENT WEEK	Summative Project Presentations & Model Submissions
Dec 21 – Jan 1	Break	<i>No Class — Winter Vacation Break</i>	—
Jan 4 – Jan 5	6.A	Classical Constructions: Copying Lines, Angles, & Parallels	Complete Geometric Constructions WS1 Portfolio
Jan 6 – Jan 7	6.B	Classical Constructions: Perpendicular & Angle Bisectors	Complete Geometric Constructions WS2 Portfolio
Jan 8 – Jan 11	Proj	CONSTRUCTIONS ARCHITECTURAL PROJECT WORK	Final Design Template Link & Drafting Checkpoints
Jan 12 – Jan 15	Rev	Semester Comprehensive Concept Alignment Modules	Study Formulated Reference Note Packets (Modules 1-6)
Jan 18	PIR	<i>No Class — Staff Professional Development Day</i>	—
Jan 19	Rev	Final Portfolio Adjustments & Blueprint Audits	In-Class Terminal Preparation Frameworks
Jan 20 – Jan 22	Final	SEMESTER 1 CUMULATIVE EXAMINATION BLOCK	Official Terminal Evaluation Blocks (Red/ Black Tracks)